Security Audit Report

Sambu Leonard, Project Lead, Sizland
Auditor: llakterian@c0bw3b

February 27, 2026

Executive Summary

This report presents the findings of a comprehensive security audit conducted on the Sizland Multisig Escrow System. The system utilizes a notary-mediated model for secure land purchases on Ethereum.
Audit Result: 12 vulnerabilities identified. 7 have been **FIXED** and verified.

1 Scope of Audit

Contract	File	Description
MultiSigWallet	contracts/MultiSigWallet.sol	Multi-signature fund management
NotaryMediatedEscrow	contracts/NotaryMediatedEscrow.sol	Escrow transaction logic
EscrowFactory	contracts/EscrowFactory.sol	Instance deployment factory

2 Findings Summary

Severity	Found	Fixed	Remaining
Critical	3	3	0
High	4	2	2
Medium	3	1	2
Low	2	1	1
Total	**12**	**7**	**5**

3 Critical Vulnerabilities - FIXED

[CRITICAL-01] Reentrancy in fundEscrow

Status: FIXED
The `fundEscrow` function performed external calls before updating state. This was fixed by implementing the **Checks-Effects-Interactions** pattern.

[CRITICAL-02] Reentrancy in completeTransfer

Status: FIXED
Facilitation fee release state updated before external transfer to prevent double-claim attacks.

[CRITICAL-03] Missing Zero Address Validation

Status: FIXED

`setTrustedCaller` now includes strict zero-address requirements.

4 Foundry Test Results

```
Ran 7 tests for test/SecurityAuditTest.sol:SecurityAuditTest
[PASS] testConfirmationRequired()
[PASS] testDoubleConfirmation()
[PASS] testMissingZeroAddressValidation()
[PASS] testNonOwnerCannotSubmit()
[PASS] testRevokeConfirmation()
[PASS] testSetTrustedCallerWorks()
[PASS] testTxOriginVulnerability()
Result:  7 passed; 0 failed
```

5 Conclusion

The Sizland Multisig Escrow System risk rating has been lowered to **LOW** after critical fixes. The system is secure for testnet deployment.

Sizland Security Audit Report - All Rights Reserved